

STUDENT NUMBER.....



***GOSFORD HIGH SCHOOL***

**2020**

**HIGHER SCHOOL CERTIFICATE**

***FINAL ASSESSMENT***

# **Year 11 PRELIMINARY MATHEMATICS – EXTENSION 1**

**Time Allowed** - 90 minutes plus 10 minutes reading time

- Write using a black or blue pen. Black pen is preferred.
- Board approved calculators may be used
- Answers to the multiple choice are to be done on the answer sheet provided.
- Start each section on a **new** page. **Each section** will be collected **separately.**
- Show all relevant mathematical reasoning and/or calculations.
- **THIS PAPER IS TO BE RE-COLLECTED AT END OF EXAM.**

## Section I

## Attempt Questions 1-8 (8 Marks)

Allow about 15 minutes for this section.

Use the multiple-choice answer sheet for questions 1-10.

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- 1 What is the Cartesian equation of the curve with parametric equations  $x = 20t$  and  $y = 40t - 5t^2$  ?

(A)  $y = 2x - \frac{x^2}{400}$

(B)  $y = 2x - \frac{x^2}{200}$

(C)  $y = 2x - \frac{x^2}{160}$

(D)  $y = 2x - \frac{x^2}{80}$

- 2 What is the domain of the function  $f(x) = \ln(3x - x^2)$  ?

1

(A)  $(0, 3)$

(B)  $[0, 3)$

(C)  $(0, 3]$

(D)  $[0, 3]$

- 3 For what value of  $k$  is the polynomial  $P(x) = kx^2 + (2k-1)x + k$  a perfect square?

1

(A)  $k = \frac{1}{9}$

(B)  $k = \frac{1}{4}$

(C)  $k = 4$

(D)  $k = 9$

- 4 Which of the following is NOT a factor of  $P(x) = x^3 + 2x^2 - 5x - 6$  ?

1

(A)  $(x+1)$

(B)  $(x-2)$

(C)  $(x-3)$

(D)  $(x+3)$

**Marks**

- 5 What is the value of  $\cos^{-1}(\tan x) + \sin^{-1}(\tan x)$  for  $-\frac{\pi}{4} \leq x \leq \frac{\pi}{4}$  ? 1
- (A) 0  
(B) 1  
(C)  $\frac{\pi}{2}$   
(D)  $\pi$
- 6 Which of the following is NOT an expression for  $\tan\left(\frac{x}{2} + \frac{\pi}{4}\right)$  ? 1
- (A)  $\sec x + \tan x$   
(B)  $\frac{1 + \sin x}{\cos x}$   
(C)  $\frac{\cos x}{1 - \sin x}$   
(D)  $(1 + \tan x)^2$
- 7 In how many ways can a group of 6 people line up in a queue so that the oldest person is in front of the youngest person? 1
- (A) 180  
(B) 360  
(C) 540  
(D) 720
- 8 What is the expansion of  $(3 - 2\sqrt{2})^3$  ? 1
- (A)  $27 - 16\sqrt{2}$   
(B)  $51 - 34\sqrt{2}$   
(C)  $99 - 70\sqrt{2}$   
(D)  $171 - 124\sqrt{2}$

## Section II

**Attempt Questions 9-12****Allow about 1 hour and 45 minutes for this section.**

Answer the questions in writing booklets provided. Use a separate writing booklet for each question. In Questions 11-14 your responses should include relevant mathematical reasoning and/or calculations.

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**Question 9 (12 Marks)****Use a separate writing booklet.**

- (a) Solve the inequality  $|2x + 1| \geq 3$  . 2
- (b) The equation  $P(x) = 0$  where  $P(x) = x^3 + cx + d$  has a double root  $x = 1$  . Find the values of  $c$  and  $d$  . 2
- (c) Show that  $\frac{1 - \cos 2x + \sin 2x}{1 + \cos 2x + \sin 2x} = \tan x$  .
- (d) Solve the inequality  $x + \frac{4}{x} \geq 5$  . 3
- (e) Consider the polynomial  $P(x) = k^2x^3 - k^3x^2 - x + k$  where  $k \neq 0$  .
- (i) Show that  $(x - k)$  is a factor of  $P(x)$  . 1
- (ii) Solve the equation  $P(x) = 0$  . 2

**Marks****Question 10 (12 Marks)****Use a separate writing booklet.**

(a) Find the number of ways that 4 students can be chosen from 6 juniors and 3 seniors

(i) without restriction. **1**

(ii) so that there are unequal numbers of juniors and seniors. **2**

(b) The equation  $2x^3 + 6x^2 + 4x + 1 = 0$  has roots  $\alpha$ ,  $\beta$  and  $\gamma$ .

(i) Find the value of  $(1 - \alpha) + (1 - \beta) + (1 - \gamma)$ . **1**

(ii) Find the value of  $(1 - \alpha)(1 - \beta)(1 - \gamma)$ . **2**

(c) Write  $\cos x - 3 \sin x + 3$  using the “ $t$ ” results given that  $t = \tan \frac{x}{2}$ . **3**

d) **1**

1. Write down the binomial expansion of  $(1 + x)^n$  **2**

11. Use the expansion of  $(1 + x)^n$  to show that  ${}^nC_0 + {}^nC_1 + \dots + {}^nC_n = 2^n$

**Marks****Question 11 (12 Marks)****Use a separate writing booklet.**

- (a) Find the multiplicity of the root  $x = -1$  of the equation  $P(x) = 0$  where  $P(x) = x^4 + 4x^3 + 7x^2 + 6x + 2$ . 2
- (b) Sketch the graph of the function  $f(x) = -2 \tan^{-1} x$  showing the intercepts on the axes and the equations of the asymptotes. 2
- (c) Solve the equation  $\sin^2 \theta = \cos \theta + 1$  where  $0 \leq \theta \leq 2\pi$ . 3
- (d) Find the centre and radius of the circle  $x^2 + y^2 - 6x + 2y + 6 = 0$  2
- (e) Consider the function  $f(x) = (x - 2)^2$ ,  $0 \leq x \leq 2$ .  
Find the inverse function  $f^{-1}(x)$  and state its domain. 3

**Question 12 (12 Marks)**

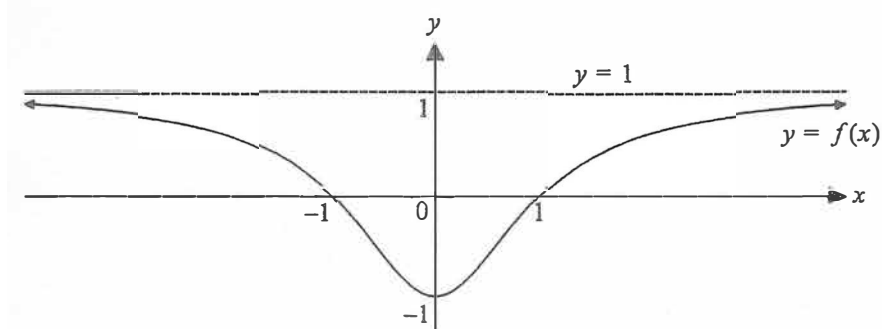
Use a separate writing booklet.

- (a) Find the number of ways in which the letters of the word TRIANGLE can be arranged in a line
- (i) so that all the vowels are next to each other and all the consonants are next to each other. 1
- (ii) so that consonants are in the two end positions. 1

- (b) If  $\sec x + \tan x = k$ , find in terms of  $k$  the value of  $\sec x$ . 2

- c) Find the number of ways in which 2 boys and 5 girls can be arranged in a circle
- i. if there is no restrictions 1.
- ii. if the boys are to sit together 1.

(d)



The diagram shows the graph of the function  $f(x) = \frac{x^2 - 1}{x^2 + 1}$ . On separate diagrams sketch the graphs of the following curves, showing clearly the intercepts on the axes and the equations of any asymptotes.

- (i)  $y = |f(x)|$ . 2
- (ii)  $y = \frac{1}{f(x)}$ . 2

- (e) i. Express  $\cos x - \sqrt{3}\sin x$  in the form  $R\cos(x + \alpha)$  where  $R > 0$  and  $0 \leq x \leq 2\pi$  2

(1/8)

# ANSWERS TO 4/11 EXT ONE PRELIM FINAL 2020

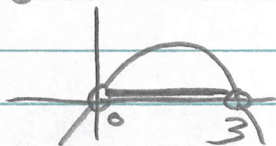
$$\begin{aligned} 1/ \quad x &= 20t \Rightarrow t = \frac{x}{20} \\ y &= 40t - 5t^2 \\ y &= 40\left(\frac{x}{20}\right) - 5\left(\frac{x^2}{20^2}\right) \end{aligned}$$

$$\begin{aligned} &= 2x - \frac{5x^2}{400} \\ &= 2x - \frac{x^2}{80} \end{aligned}$$

(D)

$$2/ \quad f(x) = \ln(3x - x^2)$$

for  $3x - x^2 > 0$



$$\therefore \text{d: } 0 < x < 3$$

ie  $(0, 3)$

(A)

$$3/ \quad P(x) = k(x^2 + (2 - \frac{1}{k})x + 1)$$

for  $x^2 + (2 - \frac{1}{k})x + 1$   
to be a perfect square

$$2 - \frac{1}{k} = -2$$

$$k = \frac{1}{4}$$

(B)

$$4/ \quad P(x) = x^3 + 2x^2 - 5x - 6$$
$$P(-1) = 0$$
$$P(2) = 0$$
$$P(3) = 24 \quad \checkmark (x-3)$$

check  $P(-3) = 0$

(C)

$$5/ \quad \text{for } -\frac{\pi}{4} < x < \frac{\pi}{4}$$

$$-1 < \tan x < 1$$

$$\text{let } \tan x = u$$

$$\therefore \cos^{-1} u + \sin^{-1} u = \frac{\pi}{2}$$

for all  $-1 < u < 1$

$\therefore$  (C)

6/ Using "t" results

$$\tan\left(\frac{x}{2} + \frac{\pi}{4}\right) = \frac{\tan \frac{x}{2} + \tan \frac{\pi}{4}}{1 - \tan \frac{x}{2} \tan \frac{\pi}{4}}$$

$$= \frac{t + 1}{1 - t}$$

$$A) \sec x + \tan x = \frac{1 + t^2 + 2t}{1 - t^2}$$

$$= \frac{(1+t)^2}{(1+t)(1-t)} = \frac{1+t}{1-t}$$

$$B) \frac{1 + \sin x}{\cos x} = \sec x + \tan x = A$$

$$C) \frac{\cos x}{1 - \sin x} = \frac{\cos x (1 + \sin x)}{(1 - \sin^2 x)}$$

$$= \frac{\cos x (1 + \sin x)}{\cos^2 x}$$

$$= \frac{1 + \sin x}{\cos x}$$

$$= \sec x + \tan x = A$$

$$D) \quad x = -\frac{\pi}{4} \Rightarrow (1 + \tan x)^2 = 0$$

But

$$\tan\left(-\frac{\pi}{8} + \frac{\pi}{4}\right) = \tan \frac{\pi}{8} > 0$$

$\therefore$  NOT A (D)

$\frac{2}{8}$

7 older person  
in front of younger

$$\frac{1}{2} \times 6! = 360$$

(B)

$$8(3 - 2\sqrt{2})^3$$

$$= 3^3 - 3 \times 3^2 \times 2\sqrt{2} + 3 \times 3 \times (2\sqrt{2})^2 - (2\sqrt{2})^3$$

$$= 27 - 54\sqrt{2} + 72 - 16\sqrt{2}$$

$$= 99 - 70\sqrt{2}$$

(C)

3/8

## Section II

Q9 a)  $|2x+1| \geq 3$

$$2x+1 \leq -3 \quad 2x+1 \geq 3 \quad (1)$$

$$2x \leq -4 \quad 2x \geq 2$$

$$x \leq -2 \quad x \geq 1 \quad (1)$$

b)  $P(x) = x^3 + cx + d$

roots at  $x=1, 1$  &  $\alpha$

$$1+1+\alpha = -\frac{b}{a} = 0$$

$$\therefore \alpha = -2$$

$$\text{from } \alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a} = c$$

$$-2 \times 1 + -2 \times 1 + 1 \times 1 = -3$$

$$c = -3$$

$$\text{from } \alpha\beta\gamma = -\frac{d}{a} = -d$$

$$d = -(1 \times 1 \times -2) = 2$$

$$\therefore c = -3$$

$$d = 2$$

c)  $\frac{1 - \cos 2x + \sin 2x}{1 + \cos 2x + \sin 2x}$

$$= \frac{2\sin^2 x + 2\sin x \cos x}{2\cos^2 x + 2\sin x \cos x} \quad (1)$$

$$= \frac{2\sin x (\sin x + \cos x)}{2\cos x (\cos x + \sin x)} \quad (1)$$

$$= \tan x$$

d)  $x + \frac{4}{x} \geq 5 \quad (x \neq 0)$

$$x^2 + 4x \geq 5x^2 \quad (1)$$

$$x^2 - 5x^2 + 4x \geq 0$$

$$x(x^2 - 5x + 4) \geq 0$$

$$x(x-4)(x-1) \geq 0 \quad (1)$$



(1)

$$0 < x \leq 1 \text{ and } x \geq 4$$

e)  $P(x) = k^2 x^3 - k^3 x^2 - x + k$   
 $k \neq 0$

$$P(x) = k^2 x^2 (x-k) - (x-k)$$
$$= (x-k)(k^2 x^2 - 1) \quad (1)$$

$$\therefore P(k) = 0$$

$$\text{or } k^5 - k^5 - k + k = 0$$

$$\therefore (x-k) \text{ is a factor} \quad (1)$$

$$\text{ii) } (x-k)(k^2 x^2 - 1) \quad (1)$$
$$= (x-k)(kx-1)(kx+1) \quad (1)$$

$$\text{for } P(x) = 0 \quad x = k \text{ or } \pm \frac{1}{k}$$

4/8

Q 10

a)  ${}^9C_4 = 126$  ①

ii)  ${}^9C_4 - {}^6C_2 \times {}^3C_2$  ①

$$= 126 - 45$$

$$= 81$$
 ①

b)  $2x^3 + 6x^2 + 4x + 1 = 0$

$\therefore (1-\alpha) + (1-\beta) + (1-\gamma)$

$$= 3 - (\alpha + \beta + \gamma)$$

$$= 3 - \left(-\frac{6}{2}\right)$$

$$= 6$$
 ①

ii) given that  $\left. \begin{matrix} 1-\alpha \\ 1-\beta \\ 1-\gamma \end{matrix} \right\}$  are roots

$\therefore 2x^3 + 6x^2 + 4x + 1 = 0$

is  $2(1-x)^3 + 6(1-x)^2 + 4(1-x) + 1 = 0$

gives a constant of 13  
with leading co-efficient of -2

$$\therefore \alpha\beta\gamma = -\frac{d}{a} = -\left(\frac{13}{-2}\right)$$

$$= 6\frac{1}{2}$$

c)  $\cos x - 3\sin x + 3$

$$\frac{1-t^2}{1+t^2} - 3\frac{2t}{1+t^2} + 3$$
 ①

$$\frac{1-t^2-6t+3+3t^2}{1+t^2}$$
 ①

$$= \frac{2t^2-6t+4}{1+t^2}$$

$$= \frac{2(t^2-3t+2)}{1+t^2}$$

$$= \frac{2(t-2)(t-1)}{1+t^2}$$
 ①

5/8

Q11 contd.

Q11 a

i)

$$(1+x)^n$$

$$= {}^n C_0 + {}^n C_1 x + {}^n C_2 x^2 + {}^n C_3 x^3 + \dots + {}^n C_n x^n$$

①

ii/ LHS let  $x=1$   
 $\therefore (1+1)^n = 2^n$

RHS  ${}^n C_0 + {}^n C_1 (1) + {}^n C_2 (1)^2 + {}^n C_3 (1)^3$   
 $+ \dots + {}^n C_n (1)^n$

$$= {}^n C_0 + {}^n C_1 + {}^n C_2 + \dots + {}^n C_n$$

Question 12

a)  $P(x) = x^4 + 4x^3 + 7x^2 + 6x + 2$

$$P(-1) = 0$$

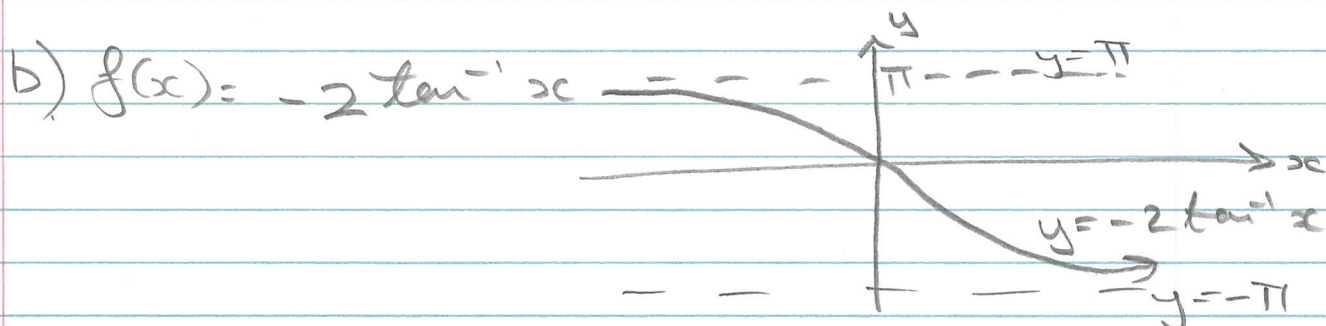
$$P'(x) = 4x^3 + 12x^2 + 14x + 6$$

$$P'(-1) = 0$$

$$P''(x) = 12x^2 + 24x + 14$$

$$P''(-1) \neq 0$$

$\therefore -1$  has a multiplicity of 2 for  $P(x)=0$



c)  $\sin^2 \theta = \cos \theta + 1$   
 $1 - \cos^2 \theta = \cos \theta + 1$   
 $\cos^2 \theta + \cos \theta = 0$   
 $\cos \theta (\cos \theta + 1) = 0$

$\cos \theta = 0$  or  $\cos \theta = -1$   
 $\theta = \frac{\pi}{2}, \frac{3\pi}{2}$  or  $\theta = \pi, 2\pi$

Check  $\frac{\pi}{2}, \frac{3\pi}{2}$  ~~and~~  $\therefore \theta = \pi$

6/8

Q12d)  $x^2 + y^2 - 6x + 2y + 6 = 0$   
 $x^2 - 6x + \underline{\quad} + y^2 + 2y + \underline{\quad} = -6$

$$x^2 - 6x + 9 + y^2 + 2y + 1 = -6 + 9 + 1$$
$$(x-3)^2 + (y+1)^2 = 4$$

centre  $(3, -1)$

radius 2

12 e)  $f(x) = (x-2)^2$

$$f(x) = y = (x-2)^2$$

$$f^{-1}(x) \cdot xc = (y-2)^2$$
$$\pm \sqrt{x} = y - 2$$

$$2 \pm \sqrt{x} = y$$

$$y = 2 \pm \sqrt{x}$$

but for  $f(x)$   $0 \leq y \leq 4$   
 $0 \leq x \leq 2$

$$y = 2 - \sqrt{x}$$

$$D: [0, 4]$$

7/8

Q 13

a) TRIANGLE

i)  $\boxed{1 \text{ AE}} \leftrightarrow \boxed{\text{TRNGL}}$

$$2 \times 3! \times 5! = 1440$$

ii)  $\begin{matrix} 1^{\text{st}} \& \text{last} & 5 \times 4 \\ \text{rest arranged} & 6! \end{matrix} \Rightarrow 5 \times 4 \times 6! = 14400$

b)  $\sec x + \tan x = k$

$$(\sec x + \tan x)(\sec x - \tan x) = \sec^2 x - \tan^2 x = 1$$

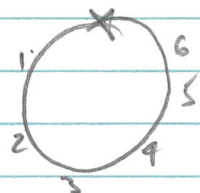
$$\therefore \sec x - \tan x = \frac{1}{\sec x + \tan x}$$

$$\text{so } \sec x + \tan x = k$$

$$\sec x - \tan x = \frac{1}{k}$$

$$2 \sec x = k + \frac{1}{k}$$

$$\sec x = \frac{1}{2} \left( k + \frac{1}{k} \right)$$

c) i)  1 fixed  $\therefore 6! = 720$

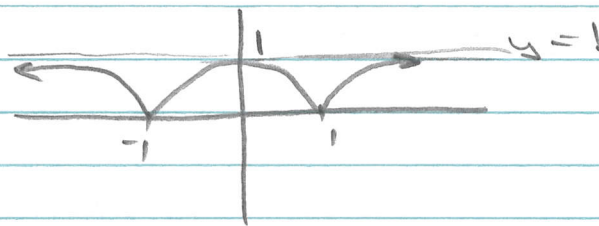
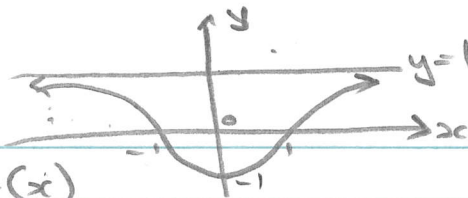
ii) Fix one girl  $\therefore 2!$  ways for boys  
 $5!$  for 4 girls and 1 boy

$$\text{Total } 5! \times 2! = 240$$

8/8

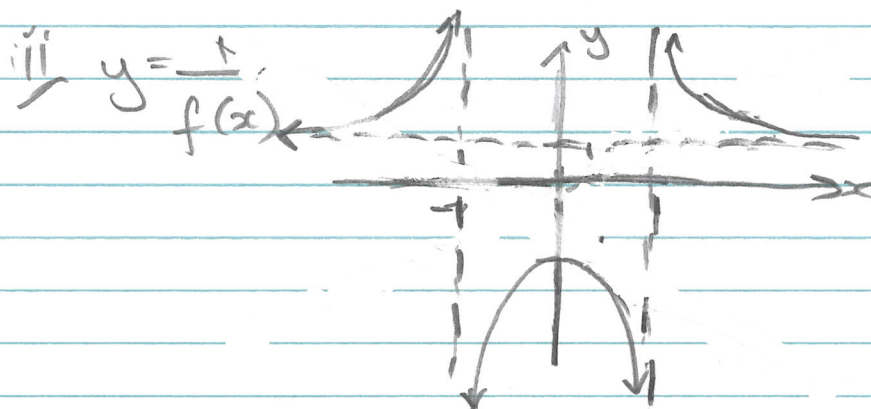
d) from

13 i)  $y = |f(x)|$



① for +ve

① for shape



① for Asymptotes

① for shape

e)  $\cos x - \sqrt{3} \sin x$

$$R \cos(x + \alpha) \equiv \cos x - \sqrt{3} \sin x$$

$$R \cos \alpha = 1$$

$$R \sin \alpha = \sqrt{3}$$

$$\therefore \tan \alpha = \sqrt{3}$$

$$\alpha = \frac{\pi}{3}$$

$$2 R^2 = 1 + 3$$

$$R^2 = 4$$

$$R = 2$$

for  $R > 0$

$$\therefore 2 \cos\left(x + \frac{\pi}{3}\right)$$